



MAIN Hub

MICROELECTRONICS FOR AMERICAN INNOVATION AND NATIONAL SECURITY

MICROELECTRONICS COMMONS

One network. A common operating system.

The current and proposed COMMONS organizational structure, the ARI network administrator (franchise) model, and two pathways to a microelectronics marketplace.

PREPARED FOR

The Honorable Mike Dodd

ASW(CT) / OUSW(R&E)



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Current org chart

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Proposed org chart

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context

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Franchise model: ARI &
services

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Franchise model: end
state

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Use the A&S
Marketplace (APEX)

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Start your own
marketplace

— TODAY

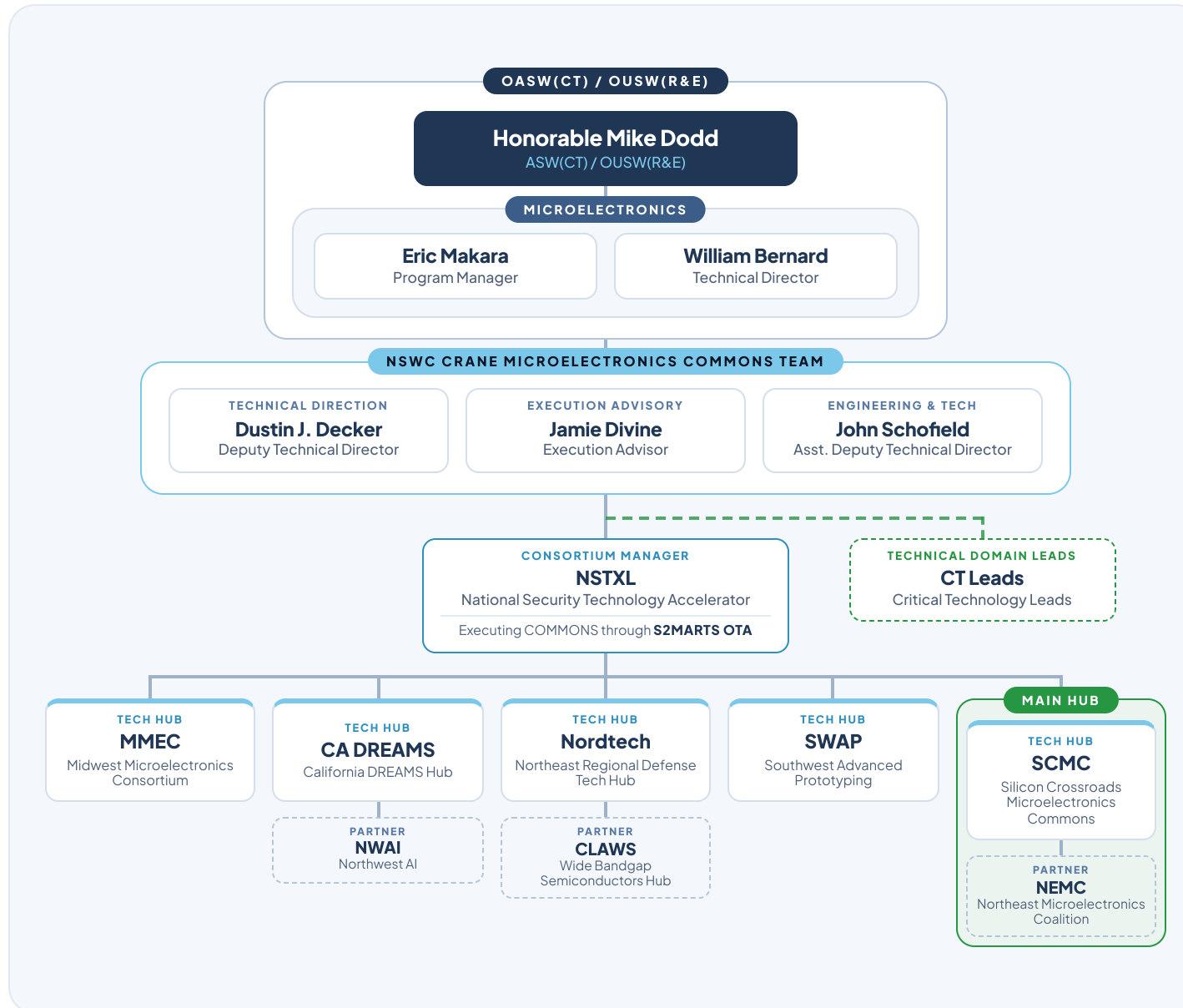
Current COMMONS organizational structure

Five Tech Hubs connect directly to the consortium manager, and each one runs its own operating model.

HOW IT WORKS TODAY

- **NSTXL** is the consortium manager and executes COMMONS through the **S2MARTS OTA**.
- **Hubs** connect directly to NSTXL as the technical execution hubs.
- **CT Leads** provide domain-specific technology governance and technical alignment as an external advisory arm.
- The Hubs were stood up **independently**, each with its own organization, leadership, relationships, capabilities, and priorities.

- Executive & program oversight
- NSWC Crane team
- Consortium manager
- CT Leads (advisory)
- Hub partner



THE RESULT

Individual Hubs



Individual operating models

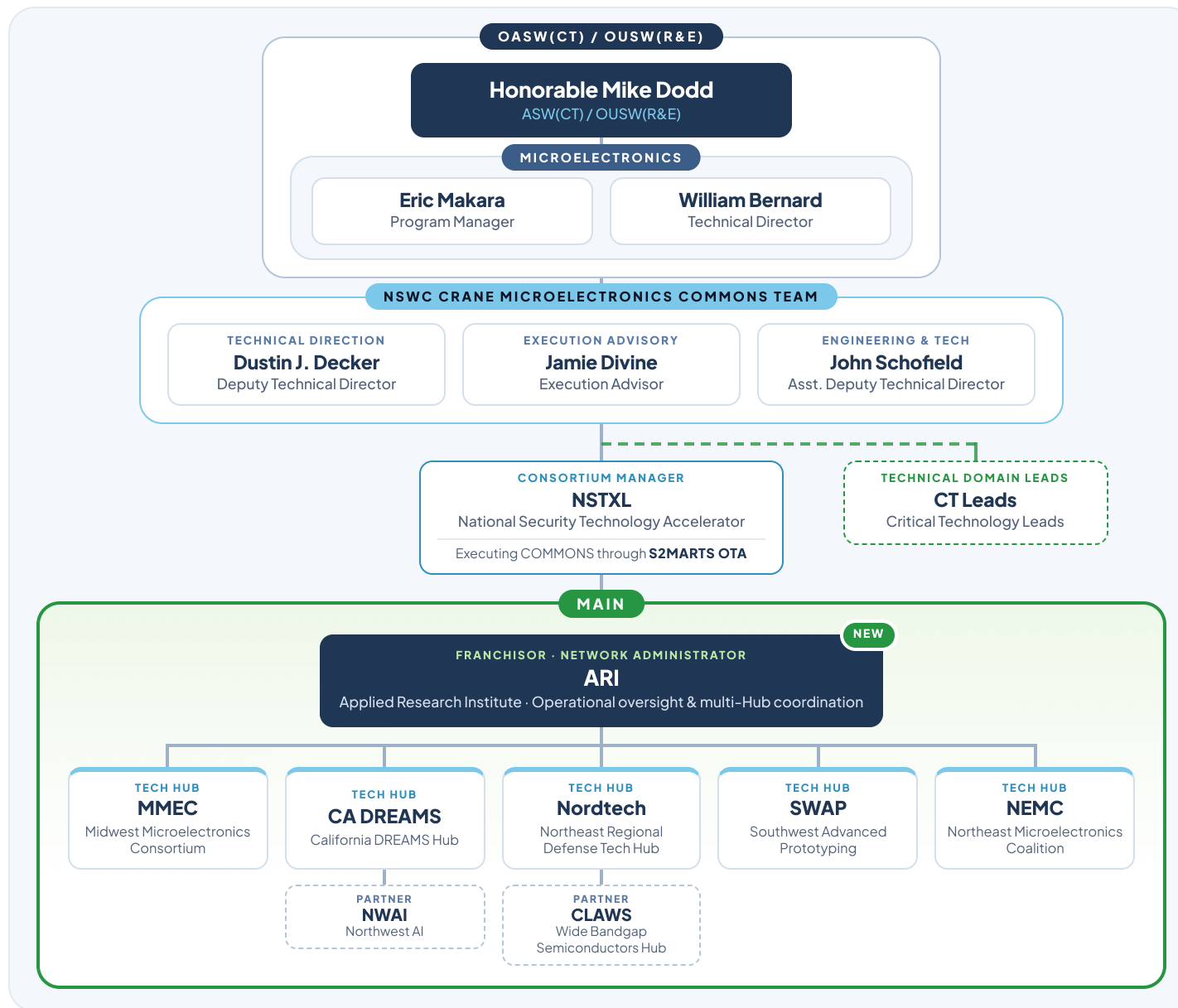


Individual systems

PROPOSED

Proposed COMMONS organizational structure

ARI becomes the network administrator (the franchisor) inside a MAIN Hub boundary, directly above all Hubs.



WHAT CHANGES

- **ARI becomes the network administrator**, providing a common operating system for every Hub inside the MAIN Hub boundary.
- **ARI dissolves its standalone Hub role.** SCMC's capabilities become a platform that supports the entire network, and **NEMC** continues as a Tech Hub.
- **Only shared functions are centralized.** Each Hub keeps its identity, leadership, expertise, membership, and relationships.

WHAT STAYS THE SAME

- Government oversight, the NSWC Crane team, NSTXL execution through S2MARTS, and CT Leads' domain support

THE RESULT

Individual Hubs



Common network platform



Shared services & infrastructure

— THE FRANCHISE MODEL

A common operating system for the Microelectronics Hubs

The Hubs grew up independently. A network-level administrator gives them one way to operate while each Hub stays independent.

Purpose and context

- 
Purpose
 Establish a common "franchise model" so the Microelectronics Hubs operate more consistently, efficiently, and in a structured way.
- 
Context
 The Hubs were created independently, each with its own organization, leadership, relationships, capabilities, and priorities.
- 
Recent lesson
 Consolidating eight Hubs into five showed how hard alignment is.

HUB CONSOLIDATION

8 → 5

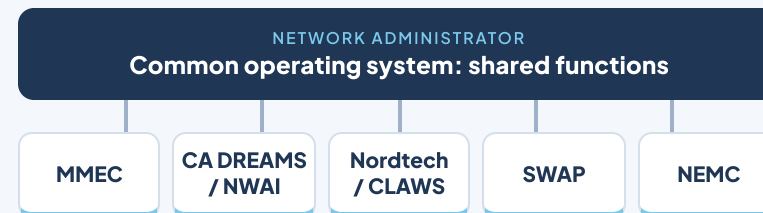
Hubs after consolidation. Getting them aligned was the hard part, and it showed that the network needs a common way of operating.

The administrator model

- 
Proposal
 Designate one organization as a network-level administrator that provides a common operating system for all Hubs.
- 
Independence preserved
 Hubs keep their identities, expertise, memberships, and relationships. Only shared functions are centralized.

Focus: centralize what benefits from consistency and scale, not the Hubs themselves.

HUBS STAY INDEPENDENT · SHARED FUNCTIONS ARE CENTRALIZED



THE FRANCHISE MODEL

Why ARI, and what it would run for the network

ARI already runs this model. It would move its SCMC capabilities out of a standalone Hub and into a platform that serves every Hub.

Why ARI fits: existing model and capabilities



Track record

ARI already coordinates through the DIU on-ramp model, providing:

- ✓ Common playbooks & SOPs
- ✓ Reporting & data
- ✓ Events & communications
- ✓ Mission & opportunity pipelines
- ✓ Validation pathways
- ✓ Resource & investment coordination



SCMC as the platform

ARI's SCMC capabilities can move from a standalone Hub to a platform that supports the entire network. ARI would dissolve its Hub role to serve as administrator.

ARI'S ROLE SHIFTS

TODAY
SCMC: a standalone Hub



PROPOSED
Platform for the entire network

Shared services under the common operating system



Operating model

Shared playbooks, SOPs, service definitions, and processes.



Data and reporting

Common metrics, pipeline visibility, and performance reporting.



Communications and events

Coordinated national programming and stakeholder engagement.



Opportunity pipeline and validation

Visibility into DoD needs and into funding and acquisition pathways, plus standardized validation.



Investment and marketplaces

Coordinated access to private capital, plus shared platforms for technologies, companies, and manufacturing capabilities.

THE FRANCHISE MODEL

Hubs stay distinct. The network goes national.

What does not change: Hub autonomy and national reach



Autonomy

No Hub is centralized. Each one keeps:

- ✓ Identity
- ✓ Leadership
- ✓ Expertise
- ✓ Membership
- ✓ Relationships
- ✓ Areas of focus



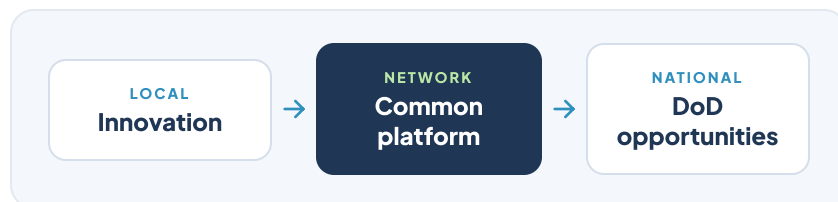
National orientation

Many Hubs already operate nationally. A network administrator reinforces that and keeps the Hubs from becoming silos.



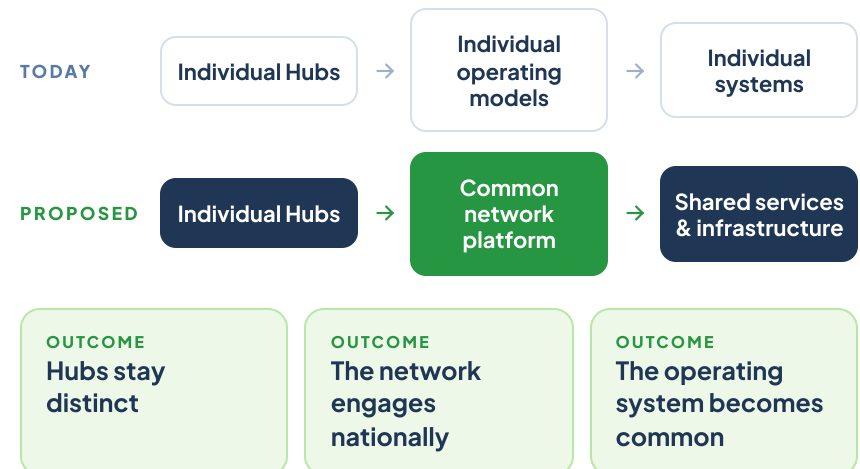
Local to national

Hubs connect local innovation strengths to national opportunities through the network.



End state and value to government

No Hub is centralized. The network gets a common operating system, and the government gets shared visibility across all of it.



VALUE TO GOVERNMENT

- Shared visibility & infrastructure
- Common processes
- Broader access to technologies & companies
- Coordinated validation
- Better investment connections
- Clearer acquisition & adoption pathways

MARKETPLACE & TIMELINES

Moving award timelines to the speed of operations

The ASW wants technology developed in 24 months or less. Our goal is to move from call to award 10× faster, cutting award time from 50% of technical execution to under 5%.

COMMONS Call for Proposals: proposal & award timeline



OPTION 1 · SEE PAGE 08

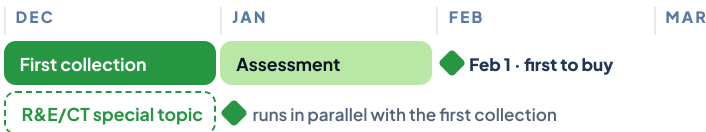
Leverage the existing A&S Marketplace (APEX)



GO LIVE
Dec 1

COST / TOPIC
~\$400K

FIRST BUY
Feb 1



APPROACH	Use APEX, where A&S already has a microelectronics category
R&E ROLE	Topic sponsor: adds funding to drive topics or shopping notices
BUYER	NSTXL could still perform the buy

~\$400K fee for a single topic. Live **December 1**, with the first buy on **February 1**.

OPTION 2 · SEE PAGE 09

Stand up a DoW Microelectronics Marketplace



GO LIVE
60 days

COST
\$3M / yr

CONTRACTING
Sole-sourceable

On contract → stand up the marketplace

Live

DAY 0 | DAY 30 | DAY 60

APPROACH	Stand up an independent, brand-new DoW marketplace
CONTRACTING	Our marketplaces are sole-sourceable, so getting on contract is straightforward
FUNDING	Requires some funding for projects to back a new marketplace

\$3M per year for a brand-new program, live **60 days** after getting on contract.

— OPTION 1 · USE THE A&S MARKETPLACE

Leverage the existing A&S Marketplace (APEX)

Build on the APEX Marketplace, where A&S already has a microelectronics category. R&E adds funding to drive topics or shopping notices.

APEX GOES LIVE

Dec 1

First collection opens

FEE PER TOPIC

~\$400K

Fee for a single topic

FIRST TO BUY

Feb 1

After January assessment

🕒 STRUCTURE & TIMELINE

1

DECEMBER 1

Go live with the first collection

APEX opens with the A&S microelectronics category. R&E funding drives topics or shopping notices.

2

JANUARY

Assessment

Submissions are assessed, with an option for a 30-day assessment window.

3

FEBRUARY 1

First to buy

First purchases can be made off the marketplace.

In parallel: R&E/CT can run a special topic alongside the first collection.

⚙️ PROCESS ADJUSTMENTS (OPTIONAL)

The submission format can be adjusted to add rigor if required:

- ✓ 5-minute video + white paper submission
- ✓ 30-day assessment window

🛒 WHO PERFORMS THE BUY

NSTXL could still perform the buy. There's precedent on Tradewinds:

MARKETPLACE
Tradewinds



SPECIAL TOPICS
WIRE



PERFORMED THE BUY
ATI

⚖️ HOW IT COMPARES WITH OPTION 2

	OPTION 1 · APEX	OPTION 2 · OWN MARKETPLACE
GO LIVE	Dec 1	60 days after getting on contract
COST	~\$400K per topic	\$3M per year
PLATFORM	Existing A&S category	Brand-new program

BOTTOM LINE

Uses the existing A&S microelectronics category on APEX. It goes live December 1 for a ~\$400K fee for a single topic, with the first buy on February 1.

— OPTION 2 · START YOUR OWN MARKETPLACE

Stand up a DoW Microelectronics Marketplace

Stand up an independent effort: a dedicated marketplace for microelectronics, live 60 days after it goes on contract.

TIME TO GO LIVE

60 days

From getting on contract

ANNUAL COST

\$3M/yr

For a brand-new program

CONTRACTING

Sole-sourceable

 OVERVIEW

Stand up an **independent DoW Microelectronics Marketplace**. Our marketplaces are sole-sourceable, so **getting on contract is straightforward**. Once on contract, it takes **60 days to go live**.

 KEY ELEMENTS

- ✓ **Project funding required:** some funding is needed for projects to back a new marketplace.
- ✓ **Annual cost:** \$3M per year for a brand-new program.
- ✓ **Timeline:** go live 60 days after getting on contract.
- ✓ **Open to all government buyers:** any government agency can purchase from the marketplace, so COMMONS can attract investment from outside the COMMONS program.

TIME TO GO LIVE

On contract → stand up the marketplace

 Live

DAY 0

DAY 30

DAY 60

 HOW IT WORKS

1

CONTRACT

Get on contract by sole source

Our marketplaces are sole-sourceable, so contracting is straightforward.

2

DAYS 0 – 60

Stand up the marketplace

Build a dedicated DoW Microelectronics Marketplace as an independent effort.

3

DAY 60

Go live

The marketplace opens for microelectronics technologies and companies.

4

ONGOING · \$3M / YR

Operate and fund projects

Run the program and fund the projects that back the new marketplace.

BOTTOM LINE

A fully independent program, contracted by sole source and live in 60 days. It carries a \$3M annual cost and needs project funding behind it.



Hubs stay **distinct.**
The network becomes **national.**
The operating system becomes **common.**

A common network platform gives the Department shared visibility and infrastructure, common processes, and broader access to technologies and companies. It also brings coordinated validation, better investment connections, and clearer acquisition and adoption pathways.



MAIN Hub

MICROELECTRONICS FOR AMERICAN INNOVATION AND NATIONAL SECURITY

Applied Research Institute

theari.us · October 2026